

# L8: Application in Climate Science

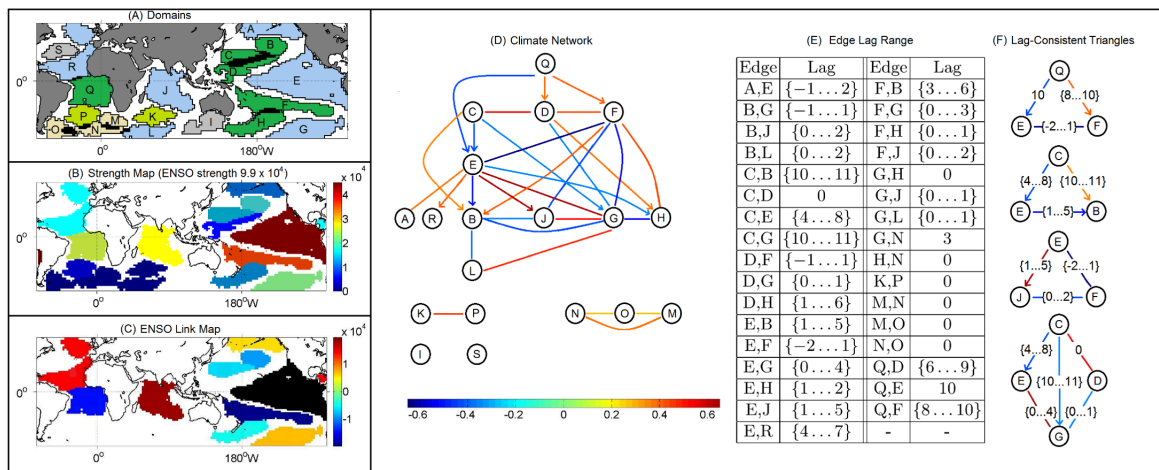


Figure from "[5-MAPS: from spatio-temporal data to a weighted and lagged network between functional domains](https://link.springer.com/article/10.1007/s41109-018-0078-z)" by Fountalis, I., Dovrolis, C., Bracco, A. et al.,

Here is a summary of the results when Delta-MAPS was applied on the Sea-Surface Temperature (SST) monthly data mentioned earlier (time period 1956-2005).

The number of grid points is 6000. Delta-MAPS identified 18 domains, showing the great potential for dimensionality reduction using this method.

35% of the grid cells do not belong to any domain.

The largest and strongest (in terms of network connections) domain is the ENSO region at the Pacific ocean (domain E – see part-A and part-B).

The connections between the ENSO domain and other domains are shown in part-C. Some of these connections are positive correlations while others are negative. For instance, ENSO is positively correlated with the domain at the Indian ocean as well as the North Atlantic, while it is negatively correlated with the domain at the South Atlantic.

The complete functional network between domains is shown in part-D. Some edges are directional, meaning that the correlations are strongest at a certain lag. These lags are shown for each edge at part-E. For instance, the edge from Q to E means that the activity at domain Q precedes the activity of domain E with a time difference of about 10 months.

The network of part-D forms a 3-4 layer hierarchy. Even though ENSO (domain E) is the most connected domain, it is not at the root of this hierarchy. Instead, region Q, west of Africa in the South Atlantic, seems to be **"ahead"** of all other domains and it can be used to predict, to some extent, the activity of other domains months in advance.

Finally, part-F shows a number of domain **"triangles"** in this network, showing that their lags are temporally consistent. For instance, Q is ahead of E by about 10 months, Q is ahead of F by about 8-10 months, while F and E do not have a clear precedence relation between them (the lag range is between -2 to +1 months).

## Food for Thought

How would you identify the correct lag between two correlated timeseries? Why do you think Delta-MAPS report a range of such lags instead of a single value?